



CERTIFICATE OF PRESENTATION

**This is to certify that
Merve Sinem KARAHAN**

attended the 3rd International Conference on Advances and Innovations in Engineering (ICAIE) held on September 25-27, 2025 by Fırat University, Elazığ, Türkiye, and orally presented the paper entitled
“Detecting Street Signs for Visually Impaired People: A Yolov8 and Federated Learning- Based Approach”

Prof. Dr. Ebru AKPINAR
Chair of ICAIE Committee